

AI-Powered Project Monitoring Tool

ML-Based Task Allocation, Duration and Delay-Risk Prediction

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The Problem I

Why Project Monitoring Needs to Change

- Only around 35% of projects are considered successful; one cause is the low maturity of the tools used to manage them [NDDF25]
- Most teams still rely on spreadsheets and manual status updates, so delays are noticed late
- Task assignment is usually based on a manager's intuition, not on skills, past performance, or current workload [ASA+24]
- Predictive features (delay risk, effort forecasting) are rare or locked behind enterprise tools

What Is the Project? I

Project Objective

- A Jira-style project monitoring tool with **trained ML models** instead of fixed rules
- **Task allocation:** recommend the best employee for a task based on skills, past performance, workload, and customer category (e.g. repeat or high-budget customers)
- **Duration prediction:** estimate how long a task will actually take
- **Delay-risk scoring:** flag tasks likely to miss their deadline
- **Real-time dashboards:** managers see employee performance and project status in one place

Why ML for Project Management? I

State of Research

- Machine learning is the dominant AI technique in project management research, and forecasting is its most common function [TDTV23]
- AI is increasingly applied to project risk management across the whole project lifecycle [NDDF25]
- Effort estimation is heavily studied, but resource allocation and activity duration have received far less attention [KO26]
- Fully autonomous scheduling is unrealistic today; automating selected subprocesses is the practical path [KO26]

System Architecture I

How the Pieces Fit Together

- **Data layer:** SQLite database as the single source of truth
- **ML layer:** scikit-learn models trained on historical task data, retrained as new tasks complete
- **Application layer:** multi-page Streamlit app — Dashboard, Tasks, Employees, Customers, ML Insights
- **Tech stack:** Python, pandas, scikit-learn, SQLite, Streamlit

Database Design I

Core Tables

- **Employees:** skills, experience level, current workload, performance history
- **Tasks:** required skills, priority, complexity, estimated hours, deadline, status, assignee
- **Customers:** category (new, repeat, high-budget), linked projects
- **Task history:** actual hours, completed on time or not, quality outcome — the training data for all ML models

ML Model 1: Task Allocation I

Who Should Do This Task?

- **Step 1 — Skill match:** cosine similarity between the task's required skills and each employee's skill profile filters suitable candidates
- **Step 2 — Suitability model:** a Random Forest classifier predicts the probability that an employee completes the task on time and with good quality
- Features: skill match, experience, past on-time rate, current workload, task priority and complexity, customer category
- Output: a **ranked list** of employees; the manager makes the final choice
- Grounded in ML-based work assignment [AHM06] and Random Forest task allocation [ASA+24]

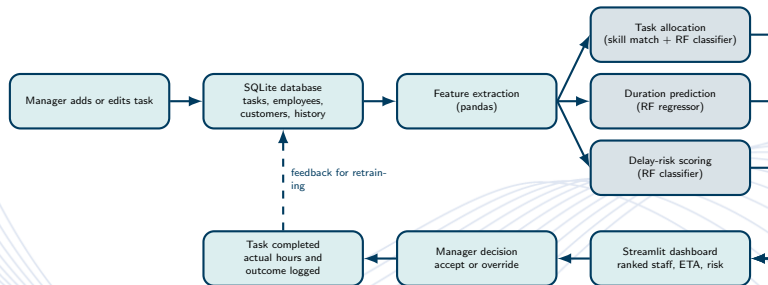
ML Models 2 and 3: Duration and Risk I

How Long Will It Take? Will It Be Late?

- **Duration prediction:** Random Forest regressor predicting actual hours from estimated hours, complexity, priority, and the assignee's experience and workload
- Tree-based models are among the most used and effective ML techniques for effort and duration estimation [KO26]
- **Delay-risk scoring:** Random Forest classifier predicting the probability that a task misses its deadline
- Risk prediction is one of the main uses of AI in project management [NDDF25, TDTV23]
- Delayed tasks are usually the minority class, so class balancing (e.g. SMOTE) will be applied [ASA+24]

System Workflow

From Task Entry to Manager Decision



Evaluation Plan I

How We Will Measure the Models

- **Data split:** 80/20 train/test split plus k-fold cross-validation [ASA+24]
- **Task allocation:** precision of the top-1 and top-3 recommendations [AHM06]
- **Duration prediction:** mean absolute error (hours) and R^2
- **Delay risk:** precision, recall, F1-score, and ROC-AUC — not accuracy alone, since classes are imbalanced
- **Baseline:** compare each model against the earlier rule-based version of the tool

Future Scope I

Beyond This Build

- Train on real Jira, Asana, or GitHub data instead of synthetic data
lack of real datasets is a known challenge in this field [KO26]
- Use NLP on task descriptions as additional features, as in text-based assignment [AHM06, ASA+24]
- Explainability: show managers *why* a task was flagged or an employee was recommended
- Richer resourcing: leave calendars and multi-project contention
- Move to a scalable backend (API layer, PostgreSQL) for multi-user use

Learnings So Far I

What This Project Has Taught Us

- Rule-based scoring is easy to build but cannot learn from past outcomes a trained model can
- Data quality and quantity decide model quality; assignment accuracy depends on having enough past tasks per person [AHM06]
- High accuracy can be misleading on imbalanced data, so F1 and AUC matter [ASA+24]
- ML should support the manager, not replace them: recommend, then let a human decide [AHM06, KO26]

References I

- [AHM06] J. Anvik, L. Hiew, and G. C. Murphy, "Who should fix this bug?," in *Proc. 28th Int. Conf. on Software Engineering (ICSE)*, Shanghai, China, 2006, pp. 361–370, doi: 10.1145/1134285.1134336.
- [ASA+24] D. Al-Fraihat, Y. Sharrah, A.-R. Al-Ghuwairi, H. Alzabut, M. Beshara, and A. Algarni, "Utilizing machine learning algorithms for task allocation in distributed agile software development," *Heliyon*, vol. 10, e39926, 2024, doi: 10.1016/j.heliyon.2024.e39926.
- [KO26] M. Koszykowski and W. Orzeszko, "Machine learning in project schedule creation: A systematic literature review," *Journal of Scheduling*, vol. 29, pp. 21–38, 2026, doi: 10.1007/s10951-025-00857-w.
- [NDDF25] M. E. Nenni, F. De Felice, C. De Luca, and A. Forcina, "How artificial intelligence will transform project management in the age of digitization: A systematic literature review," *Management Review Quarterly*, vol. 75, pp. 1669–1716, 2025, doi: 10.1007/s11301-024-00418-z.
- [TDTV23] I. Taboada, A. Daneshpajouh, N. Toledo, and T. de Vass, "Artificial intelligence enabled project management: A systematic literature review," *Applied Sciences*, vol. 13, no. 8, p. 5014, 2023, doi: 10.3390/app13085014.